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(a)	Phase difference between two waves is the fraction of a cycle by which two waves are out of step with each other.
(b)(i)	Lamp is not a point source
	Single slit makes light from S_1 and S_2 coherent / light from lamp is not coherent
(b)(ii)	One line from S_0 to S_1 to C One line from S_0 to S_2 to C Arrows to indicate direction
(b)(iii)	Path difference = 0 m
(b)(iv)1.	Intensity at C due to S_2 is reduced
	Resultant amplitude at C is less than A since intensity is proportional to amplitude ² .
(b)(iv)2.	Fringe separation is smaller
	Contrast remains the same
(c)(i)	Two waves travelling in opposite directions meet/superpose
	Two waves have same speed and frequency/wavelength

(c)(ii)1.	Correct stationary wave pattern: two nodes and two antinodes
	Pressure variation is maximum at the displacement node
	Microphone is placed 30 cm from the open end or at the closed end
(c)(ii)2.	Fundamental mode requires wavelength of 360 cm
	No stationary wave pattern can be formed
(d)	Diagram shows: Laser Grating Screen showing zero and first order bright fringe or other order fringes (can be markings on screen) Angle θ between zero and first order bright fringe
	d is slit separation or example calculation like 1 mm / 600 lines
	D is screen distance and x is distance from $n = 0$ to $n = 1$
	$\tan \theta = x / D$ To find λ , use $\lambda = d \sin \theta$